

CASTLEDERG, United Kingdom

WMO: 039040

Lat: 54.7071N

Lon: 7.5775W

Elev: 164

StdP: 14.61

Time Zone: 0.00 (EUW)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 22.5	(c) 26.5	(d) 20.6	(e) 15.5	(f) 24.2	(g) 24.3	(h) 18.5	(i) 29.2	(j) 33.6	(k) 47.4	(l) 30.1	(m) 48.0	(n) 1.8	(o) 290	(p) 0.614

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.0	(c) 72.9	(d) 63.4	(e) 69.6	(f) 61.7	(g) 66.9	(h) 60.2	(i) 65.0	(j) 70.4	(k) 63.1	(l) 67.5	(m) 61.6	(n) 65.5	(o) 8.2	(p) 130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.9	86.2	67.4	61.3	81.4	65.4	59.8	77.3	63.7	30.0	70.5	28.7	67.6	27.6	65.7	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 25.4	(b) 21.7	(c) 18.9	(e) 18.3	(f) 80.5	(g) 6.1	(h) 3.6	(i) 13.9	(j) 83.1	(k) 10.3	(l) 85.2	(m) 6.9	(n) 87.3	(o) 2.5	(p) 89.9		
(a) 25.4	(b) 21.7	(c) 18.9	(e) 18.1	(f) 69.7	(g) 5.8	(h) 3.2	(i) 13.9	(j) 72.0	(k) 10.5	(l) 73.9	(m) 7.2	(n) 75.7	(o) 3.0	(p) 78.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.6	40.4	41.1	42.9	46.1	51.2	55.9	58.4	57.7	54.7	49.5	44.1	40.7
	DBStd	8.28	5.77	5.24	4.91	4.74	4.81	4.44	4.09	3.47	4.36	5.03	5.29	7.01
	HDD50	1515	300	251	224	135	42	4	0	0	8	71	187	293
	HDD65	5996	764	668	684	566	429	275	209	228	309	481	628	755
	CDD50	1005	1	3	5	19	78	181	262	238	150	56	9	3
	CDD65	7	0	0	0	0	0	1	5	1	0	0	0	0
	CDH74	83	0	0	0	0	4	25	49	2	3	0	0	0
	CDH80	8	0	0	0	0	0	4	4	0	0	0	0	0
Wind	WSAvg	7.9	9.3	9.2	8.9	8.0	7.8	7.1	6.6	6.9	7.1	7.6	7.8	8.6
Precipitation	PrecAvg	47.7	5.1	4.0	3.2	3.0	3.4	3.1	3.3	3.7	3.6	4.8	5.2	5.4
	PrecMax	62.0	8.2	8.1	6.6	4.8	5.9	5.6	5.9	6.6	6.8	9.0	9.8	11.1
	PrecMin	36.0	0.7	1.3	1.0	1.2	0.4	1.4	1.4	0.8	0.3	1.2	2.0	1.7
	PrecStd	6.9	2.1	2.0	1.3	1.0	1.5	1.2	1.3	1.3	1.7	2.2	1.9	2.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.8	55.4	59.9	66.2	74.1	77.1	79.7	72.2	70.9	63.0	58.2	55.9
		MCWB	52.4	52.3	52.0	55.8	63.1	63.7	68.1	63.6	63.4	59.5	55.5	53.7
	2%	DB	52.6	53.1	55.4	61.4	68.6	71.9	73.5	68.5	66.9	60.4	55.7	53.7
		MCWB	50.6	50.6	50.5	53.5	59.6	62.3	64.8	61.9	60.9	57.4	54.0	51.9
	5%	DB	50.9	51.3	52.9	58.0	64.2	68.2	69.4	66.1	64.3	58.9	53.9	52.1
		MCWB	49.2	49.2	49.1	51.9	56.6	60.6	62.2	60.5	59.9	56.2	52.1	50.4
	10%	DB	49.2	49.5	51.1	55.2	60.6	64.6	66.1	64.1	62.1	57.2	52.3	50.1
		MCWB	47.5	47.4	47.8	50.4	54.2	58.4	60.3	59.3	58.2	54.6	50.4	48.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.0	53.1	53.6	57.7	64.2	66.1	69.3	65.5	64.8	60.9	56.2	54.2
		MCDB	54.3	54.5	57.1	62.6	71.7	74.5	77.8	69.6	68.1	62.2	57.3	55.5
	2%	WB	51.1	51.3	51.7	54.9	60.4	63.5	65.8	63.4	62.4	58.5	54.2	52.2
		MCDB	52.3	52.7	54.4	59.3	67.4	69.9	72.1	66.7	65.2	59.7	55.3	53.5
	5%	WB	49.5	49.6	50.1	52.8	57.7	61.2	63.2	62.0	60.6	56.7	52.4	50.7
		MCDB	50.9	51.1	52.4	57.0	62.6	66.6	67.8	65.0	63.3	58.4	53.7	52.0
	10%	WB	47.6	47.8	48.4	51.0	55.4	59.1	61.3	60.4	58.9	55.1	50.7	48.6
		MCDB	49.1	49.5	50.6	54.8	59.5	63.6	65.1	63.3	61.4	56.9	52.2	50.1
Mean Daily Temperature Range	5% DB	MDBR	9.6	10.9	13.3	15.5	15.9	14.9	14.0	13.4	13.8	12.5	10.5	9.9
		MCDBR	10.0	10.4	15.2	20.0	22.5	21.5	19.5	15.8	16.0	12.3	10.3	10.5
		MCWBR	9.7	9.1	11.0	13.2	14.6	12.9	11.8	10.4	10.8	9.8	9.6	10.1
	5% WB	MCDBR	10.2	10.1	13.2	17.5	19.6	18.8	17.6	13.7	13.9	10.9	10.4	10.6
		MCWBR	9.9	9.4	10.5	12.4	13.7	12.2	11.5	10.2	10.6	9.6	10.1	10.3
Clear-Sky Solar Irradiance	taub		0.344	0.351	0.374	0.384	0.391	0.378	0.380	0.365	0.361	0.348	0.334	0.332
	taud		2.193	2.271	2.253	2.268	2.291	2.364	2.363	2.421	2.426	2.385	2.261	2.164
	Ebn at Noon		178	227	251	266	271	276	272	270	255	228	182	152
	Edh at Noon		20	26	33	37	38	36	36	32	28	23	19	17
All-Sky Solar Radiation	RadAvg		181	397	726	1178	1496	1524	1394	1139	818	467	223	133
	RadStd		19	35	70	110	134	152	113	94	43	42	23	13

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page